

Multimodal histopathologic models stratify hormone receptor-positive early breast cancer

Corresponding Author: Professor Sohrab Shah

This manuscript has been previously reviewed at another journal. This document only contains information relating to versions considered at Nature Communications.

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

I appreciate the authors' detailed rebuttal and the interdisciplinary expertise they bring to this work, encompassing biostatistics, computational oncology, statistical learning, and computer science. While some theoretical questions have been addressed, there are still critical areas where additional clarity in results presentation and further data analysis would be beneficial, especially regarding diagnostic metrics for clinical applicability.

The main unresolved issue is the authors' reliance on a correlation coefficient of 0.6 as a measure of acceptable predictive accuracy, as shown in Figure 2. They also reference an ROC > 0.8 (Figure 2) to indicate the tool's robustness in predicting high-risk patients (Figure 3). However, no data explicitly shows the tool's performance at assigning risk at the individual sample level compared to a gold-standard RS score. Without such data, it is unclear if this tool can reliably predict individual patient risk groups—an essential requirement for clinical application. Providing single-sample prediction data, particularly with a focus on a 5-year prediction window, could substantiate the tool's diagnostic capability.

It is essential to emphasize that correlation does not imply causative prediction. An overall correlation of 0.6 does not indicate reliable individual outcome prediction. With respect to Figure 2, the ROC analysis appears to evaluate Orpheus' continuous score against the RS-defined risk group. However, the authors do not clarify the specific "cut-off" in Orpheus used to classify high-risk cases or whether this cut-off varies across populations. Indicating whether the Orpheus cut-off is consistent across cohorts or population-specific would significantly inform its reliability as a clinical tool.

Given the claimed predictive accuracy of 0.80, demonstrating predictive performance at the single-sample level and clarifying whether a consistent cut-off applies across different cohorts would be helpful. Presenting the high-risk vs. low-risk classification from Orpheus and directly comparing it with the RS-defined high-risk group would offer a clearer picture of the tool's predictive accuracy.

Furthermore, the authors' assertion that Orpheus enhances risk stratification beyond the Oncotype DX RS (Figure 3) would benefit from some tempering. From page 8, it seems that linear models were used to develop Orpheus for risk inference within Oncotype-defined risk groups. However, it is not clear if the comparison of Orpheus with Oncotype RS, both within the training set and across external datasets, provides an equitable evaluation of predictive power.

A more robust assessment of the tool's clinical predictive power could involve a calibration plot that compares predicted probabilities (from Orpheus) with observed outcomes, ideally accounting for censoring events. This approach would better capture time-to-event dynamics, avoiding the limitations of a simple dichotomy of distant vs. non-distant recurrence in the time-dependent ROC. A 5-year prediction window, in particular, would provide valuable insight into the tool's potential in a clinical setting.

(Remarks on code availability)

Haven't reviewed the code

Version 1:

Reviewer comments:

Reviewer #4

(Remarks to the Author)

The authors have addressed my comments. I do not have further comment.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

RESPONSE TO REVIEWERS' COMMENTS

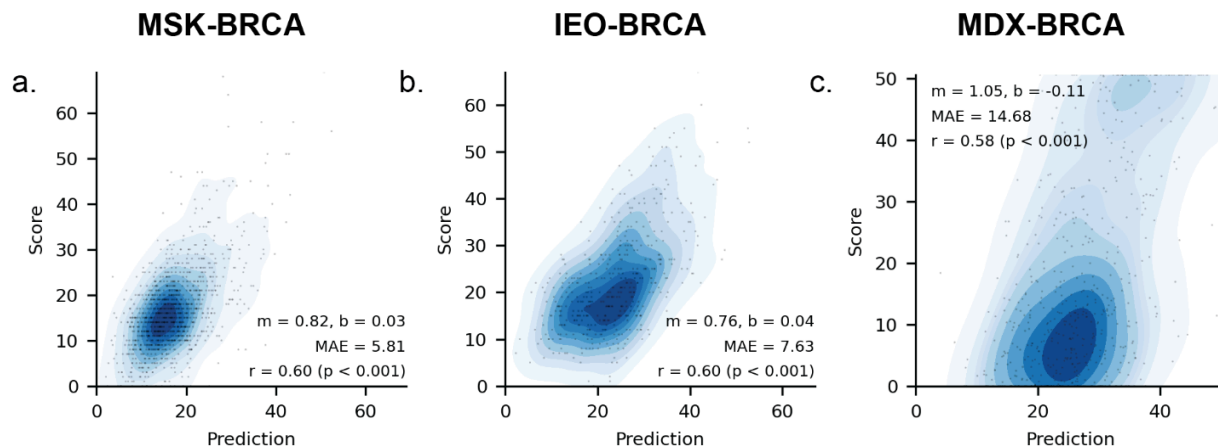
I appreciate the authors' detailed rebuttal and the interdisciplinary expertise they bring to this work, encompassing biostatistics, computational oncology, statistical learning, and computer science. While some theoretical questions have been addressed, there are still critical areas where additional clarity in results presentation and further data analysis would be beneficial, especially regarding diagnostic metrics for clinical applicability.

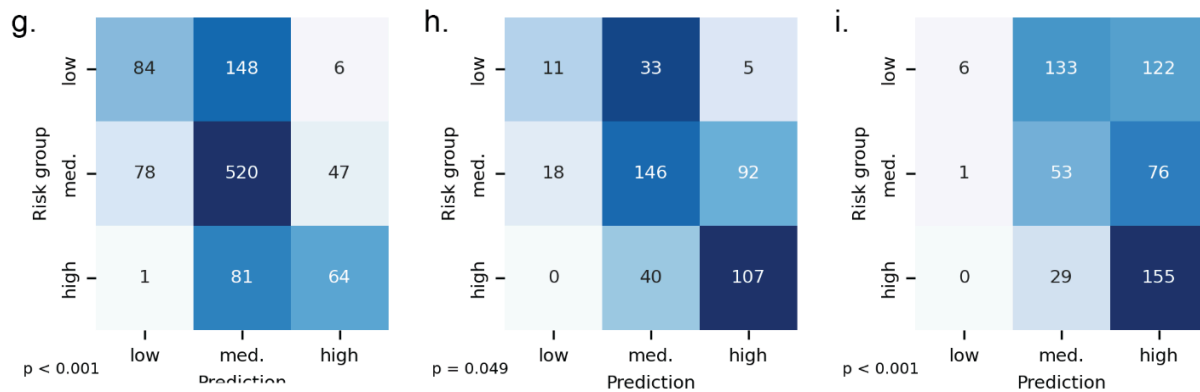
Authors' Reply: Thank you for the positive feedback. We have clarified the results and methods in the manuscript, and show additional data on Orpheus' prognostic capabilities, see the point-by-point response below.

The main unresolved issue is the authors' reliance on a correlation coefficient of 0.6 as a measure of acceptable predictive accuracy, as shown in Figure 2. They also reference an ROC > 0.8 (Figure 2) to indicate the tool's robustness in predicting high-risk patients (Figure 3). However, no data explicitly shows the tool's performance at assigning risk at the individual sample level compared to a gold-standard RS score. Without such data, it is unclear if this tool can reliably predict individual patient risk groups—an essential requirement for clinical application. Providing single-sample prediction data, particularly with a focus on a 5-year prediction window, could substantiate the tool's diagnostic capability.

Authors' Reply: Thank you for your comments. Our validation approach extends well beyond the 0.6 correlation coefficient, encompassing multiple evaluation metrics which indicate a robust and consistent signal in our 3 large cohorts. We now provide a supplementary table (Supp. Tab. 8) that includes the patient-level granularity. The scatter plots and confusion matrices (Ext. Fig. 2a-c,g-i; 3a-b,g-h; 4a-c,j-l; 5a,e) summarize this table and directly address individual sample-level risk assessment by providing point-by-point comparisons between our model's predictions (x-axis, 0-100) and the gold-standard RS scores (y-axis, 0-100), enabling evaluation of inference

accuracy for each individual sample as a visualization of the included figure source data (see figures below). Additionally, we demonstrated the model's classification metrics based on the TAILORx ODX RS thresholds and time-dependent AUROCs for distant metastatic recurrence prediction. We have enhanced clarity on these points in the methods and have comprehensive individual sample data prepared for inclusion in the source materials accompanying these figures. Specifically, we added the following to the model evaluation methods section: *“Summarizing, we evaluated Orpheus through a two-pronged approach. First, we assessed its ability to predict ODX RS values by comparing sample-level continuous risk predictions using Pearson correlation and concordance correlation coefficient. We then evaluated its performance in classifying patients into TAILORx risk groups, measuring agreement with the gold-standard RS categorization through multiple metrics: AUROC, AUPRC, F1 score, accuracy, Cohen's Kappa, and Matthew's Correlation Coefficient. In the second step, we compared both Orpheus and RS against actual clinical outcomes, specifically distant metastatic recurrence. This comparison utilized calibration plots and time-dependent AUROCs which account for censoring events to capture time-to-event dynamics beyond the clinically relevant 5-year prediction window.”*





It is essential to emphasize that correlation does not imply causative prediction. An overall correlation of 0.6 does not indicate reliable individual outcome prediction. With respect to Figure 2, the ROC analysis appears to evaluate Orpheus' continuous score against the RS-defined risk group. However, the authors do not clarify the specific “cut-off” in Orpheus used to classify high-risk cases or whether this cut-off varies across populations. Indicating whether the Orpheus cut-off is consistent across cohorts or population-specific would significantly inform its reliability as a clinical tool.

Authors' Reply: Thank you for your comment. We agree that correlation does not imply causation. Per the TAILORx-defined high-risk group, we have applied a cutoff score of 25 in all analyses (c.f. **Methods p. 8**: “Areas under the precision recall and receiver operating characteristic curve were calculated using binary thresholding of high- and low-risk disease. Unless otherwise specified, we consistently define high-risk as scores > 25 and low-risk as scores ≤ 25, following TAILORx thresholds. This standardized cut-off applies to both Orpheus and OncotypeDX recurrence scores, independent of age or nodal status, throughout the entire manuscript for direct comparability between the

*predicted Orpheus scores and OncotypeDX recurrence scores.”, **sentence 4 of the summary**: “Our model identifies TAILORx high-risk cases ($RS > 25$) with an area under the curve (AUC) of 0.89, compared to a leading clinicopathologic nomogram with 0.73”, and **main text p. 5**: “A patient is considered high-risk with a molecular $RS > 25$, following TAILORx.”). We clarified on p. 12 of the main text that correlation does not specify outcome prediction: “We further show that this correlation with RS corresponds to meaningful prognostication” to separate the correlation with ODX RS values from inference of distant recurrence outcomes. Source data for time-dependent ROC curves are available to accompany the figures. The high-risk group threshold, per TAILORx, is consistently applied as >25 in all our analyses except Supp Tab 6 where we test other thresholds. All figures inferring TAILORx high-risk disease use this threshold, which we have further clarified in the legend of Figure 2.*

Given the claimed predictive accuracy of 0.80, demonstrating predictive performance at the single-sample level and clarifying whether a consistent cut-off applies across different cohorts would be helpful. Presenting the high-risk vs. low-risk classification from Orpheus and directly comparing it with the RS-defined high-risk group would offer a clearer picture of the tool’s predictive accuracy.

Authors’ Reply: Thank you for your feedback. We maintain a consistent cut-off value of 25 for high-risk vs. low-risk classification throughout our analysis, as evidenced in all figures, main text results, and extended data figures, for both the Orpheus prediction and the gold-standard RS (c.f., Methods p. 8, sentence 3 of the summary, and main text p. 5). The only exception appears in Suppl. Table 6 where we evaluated various thresholds. Importantly, whenever Orpheus and RS scores were compared, identical RS-defined cut-off values were used to ensure fair comparison. The comparison of the Orpheus score against the RS is found in **Fig. 2a-g**.

To further corroborate our findings, we evaluated the distant recurrence-free survival by the time-dependent ROC method as defined by both RS and our tuned

Orpheus+ model (**Fig. 3c-d, Ext Data Fig 8**). The results demonstrate strong concordance in inferring distant recurrences, underlining the tool's accurate prognostic capabilities.

Furthermore, the authors' assertion that Orpheus enhances risk stratification beyond the Oncotype DX RS (Figure 3) would benefit from some tempering. From page 8, it seems that linear models were used to develop Orpheus for risk inference within Oncotype-defined risk groups. However, it is not clear if the comparison of Orpheus with Oncotype RS, both within the training set and across external datasets, provides an equitable evaluation of predictive power.

Authors' Reply: We agree with some tempering of this claim and have added text accordingly to the manuscript to clarify that we extend the *prognostic* value but not the predictive value. We have also tempered this by adding text that further external testing (beyond the two external test sets we have). The text on page 12 now reads *"Furthermore, for patients who are treated per TAILORx with adjuvant chemoendocrine or endocrine therapy based on ODX RS results, Orpheus identifies distant metastatic recurrences more accurately than ODX RS itself in the test set. With further validation, this prognostic value has the potential to refine patient selection for personalization of the adjuvant treatments and follow-up strategies."* **Extended Data Figure 8** shows further comparisons in just the test set (no validation set) and including cases with any Oncotype DX recurrence score (no thresholding), and the improvement in performance noted in **Figure 3** is preserved.

A more robust assessment of the tool's clinical predictive power could involve a calibration plot that compares predicted probabilities (from Orpheus) with observed outcomes, ideally accounting for censoring events. This approach would better capture time-to-event dynamics, avoiding the limitations of a simple dichotomy of distant vs. non-distant recurrence in the time-dependent ROC. A 5-year prediction

window, in particular, would provide valuable insight into the tool's potential in a clinical setting.

Authors' Reply: Our manuscript already contained a calibration plot comparing Orpheus predictions with observed outcomes, accounting for censoring events to capture time-to-event dynamics using the scikit-surv function [cumulative_dynamic_auc](#), which incorporates a 5-year prediction window. We have updated our methods on outcome modeling to clarify this. Specifically, we added the following: *“Calibration curves were plotted using the corresponding scikit-learn functions, comparing predicted probabilities from Orpheus with observed outcomes, accounting for censoring events. Moreover, time-dependent area under the receiver operating characteristic (AUROC) curves were plotted using scikit-survival with three-month bins from 24 to 96 months, accounting for censoring events to capture time-to-event dynamics beyond the clinically relevant 5-year prediction window.”*

